LECTURE NOTES 4: TRUNKING AND INTER-VLAN ROUTING

1. UNDERSTANDING TRUNKING PROTOCOLS (802.1Q)

- ✓ ***Trunking*** is a method of carrying multiple VLANs over a single network link. This allows for efficient use of network resources and simplifies the process of routing traffic between different VLANs.
- ✓ ***VLANs (Virtual Local Area Networks):*** VLANs segment a large network into smaller, isolated networks. Devices in one VLAN can't communicate with devices in another VLAN without special routing.

Trunk Link:

- ✓ A trunk link carries traffic from multiple VLANs between switches or other network devices.
- ✓ VLAN traffic on a trunk link is tagged with a unique identifier to differentiate between the VLANs.

802.1Q Trunking Protocol:

- ✓ ***802.1Q*** is the industry standard protocol for VLAN tagging. It allows VLANs to be transported over a single physical link by inserting a 4-byte VLAN tag into Ethernet frames.
- ✓ The tag includes a VLAN ID, which specifies the VLAN to which the traffic belongs.

Key features:

- ✓ It supports VLAN tagging, which marks the frames with the correct VLAN ID as they move across the network.
- ✓ The Native VLAN is used for traffic that is not tagged (default VLAN). Traffic on this VLAN is transmitted untagged.

2. CONFIGURING TRUNK PORTS

- ✓ To allow multiple VLANs to be carried across a single physical connection between switches, a trunk port must be configured.

Trunk Port Configuration (Basic Steps):

- ✓ Identify the switch ports that will serve as the trunk link between switches or routers.
- ✓ Configure those ports to operate in trunk mode to carry traffic for all VLANs or a subset of VLANs.

Configure Trunk Port (on a Cisco switch):

```
Switch> enable
Switch# configure terminal
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# exit
```

Explanation:

- ✓ switchport mode trunk: Sets the interface to trunk mode.
- ✓ switchport trunk allowed vlan [vlan_list]: Specifies which VLANs are allowed on the trunk link.

3. INTER-VLAN ROUTING CONCEPTS

- ✓ Inter-VLAN routing allows communication between different VLANs. Since VLANs create separate broadcast domains, devices on one VLAN cannot communicate with devices on another VLAN without routing.

Two Approaches to Inter-VLAN Routing:

Router-on-a-stick (Using a Router):

- ✓ This method uses a router with a single interface connected to the switch.
- ✓ The router's single physical interface is divided into subinterfaces, each assigned to a different VLAN.
- ✓ The router performs the routing between VLANs.

Router-on-a-stick configuration:

- ✓ Each subinterface on the router is configured with an IP address and VLAN tagging using 802.1Q.

Advantages:

- ✓ Simple and works with Layer 2 switches.
- ✓ Cost-effective for small networks.

Disadvantages:

- ✓ Performance may be limited due to the single physical interface.
- ✓ Not scalable for large networks.

Layer 3 Switch (Multi-Layer Switch):

- ✓ A Layer 3 switch combines the functionality of a switch and a router. It performs routing between VLANs directly within the switch itself.
- ✓ The switch routes traffic using Switch Virtual Interfaces (SVIs), which act as the default gateway for each VLAN.

Advantages:

- ✓ Higher performance than Router-on-a-stick, as traffic doesn't leave the switch.
- ✓ Scalable for larger networks.
- ✓ Easier to manage and reduces the need for additional routers.

4. CONFIGURING INTER-VLAN ROUTING ON SWITCHES AND ROUTERS

A. INTER-VLAN ROUTING USING A ROUTER (ROUTER-ON-A-STICK):

Step-by-Step Configuration:

Create VLANs on the switch:

```
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config)# vlan 20
Switch(config-vlan)# name IT
```

Configure Trunk Port on the switch (connecting to the router):

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
```

Configure Subinterfaces on the Router for each VLAN:

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Test connectivity:

- ✓ Assign IP addresses to the devices in VLANs 10 and 20 and ping between them to verify the router is forwarding traffic between VLANs.

B. INTER-VLAN ROUTING USING A LAYER 3 SWITCH:

Step-by-Step Configuration:
Create VLANs on the switch:

```
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config)# vlan 20
Switch(config-vlan)# name IT
```

Assign Ports to VLANs (optional, for connected devices):

```
Switch(config)# interface fastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

Enable IP Routing on the Layer 3 switch:

```
Switch(config)# ip routing
```

Create Switch Virtual Interfaces (SVIs) for each VLAN:

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
```

Test Connectivity:

- ✓ Configure IP addresses for PCs in each VLAN.
- ✓ Test by pinging between devices in different VLANs.